Lecture 12 Scribe

L12_S2_A.pdf
PDF

"Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Lecture 12 slides). The target audience is a student preparing for a closed-notes exam.

Instructions:

Reason Step-by-Step: While writing the scribe, explicitly explain the reasoning behind definitions, assumptions, proofs, and example solutions exactly as presented in the provided context. Include this reasoning directly in the scribe text to help the student understand 'why' and 'how'.

Faithfulness: Include definitions, notation, theorem statements, and worked examples exactly as they appear in the slides. Do not hallucinate or add information from outside the context.

Logical Flow: Make the logical dependencies between concepts clear.

Structure: Organize the content with clear headings and subheadings corresponding to the lecture flow.

Goal: The output should allow a student to recall definitions, reconstruct proofs, and follow example solutions step-by-step."

Below is a **fully faithful lecture scribe for Lecture 12 — Transformation of Random Variables**, written strictly from the provided slides, with **step-by-step reasoning included inside the scribe**, organized in lecture order so that a student can **reconstruct formulas, derivations, and examples in a closed-notes exam**.
All statements come from the uploaded slides.

Lecture 12 Scribe

Transformation of Random Variables + Function of Two Random Variables + Example

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1. Outline of the Lecture

The lecture covers three main topics:

1. **Transformation of Random Variables**
   → Learn techniques to find the PDF of a transformed random variable.
2. **Function of Two Random Variables**
   → Joint transformations and derived distributions.
3. **Illustrative Example**
   → Detailed derivation for

$$Z = X + Y$$

Key objective stated in slides:

> How to find PDF of new transformed random variable $f_Y(y)$ when transformation is applied.
> Assumption: PDF of original RV is known.
> (page 2)

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2. Transformation of One Random Variable

We are given:

$$Y = g(X)$$

Goal:

$$f_Y(y) = ?$$

Assumption:

- $f_X(x)$ known
- $g(x)$ monotonic (important for invertibility)

Reasoning in slides:

To find PDF of transformed RV:

Step-1: Find CDF of Y

$$F_Y(y) = P(Y \leqslant y)$$

Since $Y = g(X)$,

$$F_Y(y) = P(g(X) \leqslant y)$$

If g is monotonic increasing:

$$F_Y(y) = P(X \leqslant g^{-1}(y))$$
$$= F_X(g^{-1}(y))$$

Step-2: Differentiate

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$$

Final result (slide boxed result):

$$f_Y(y) = \frac{f_X(x)}{\left| \frac{dy}{dx} \right|} \quad x = g^{-1}(y)$$

(page 3–4)

Reasoning:

- Use CDF because inequality easier to transform
- Then differentiate to get PDF
- Absolute value appears due to derivative sign

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3. Case when g(x) is decreasing

Slides show:

If g is decreasing:

$$F_Y(y) = P(Y \leqslant y) = P(X \geqslant g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiate → same final formula

$$f_Y(y) = \frac{f_X(x)}{\left|\frac{dy}{dx}\right|}$$

Change limits carefully.

(page 4)

Reasoning:

- Direction of inequality changes
- But derivative formula stays same

---

4. Example — Nonlinear Transformation

Given:

$$X \sim U(-1, 1)$$

$$f_X(x) = \begin{cases} 1/2, & -1 < x < 1 \\ 0, & otherwise \end{cases}$$

Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Find $f_Y(y)$

Step-1: Invert

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step-2: derivative

$$\frac{dx}{dy} = \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

Step-3: substitute

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$$

$$= \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

$$= \frac{1}{\pi\sqrt{1-y^2}}$$

for

$$-1 < y < 1$$

(page 5)

Reasoning:

- Need inverse function
- Need derivative of inverse
- Plug into transformation formula

---

5. Function of Two Random Variables

Now:

$$Z = X + Y$$

Goal:

$$f_Z(z) = ?$$

Slides say:

Need joint PDF

$$f_{XY}(x, y)$$

Find CDF:

$$F_Z(z) = P(Z \leqslant z) = P(X + Y \leqslant z)$$

Then differentiate.

(page 6–7)

Reasoning:

- Cannot directly differentiate without CDF

- Region of integration depends on z

---

6. Using Integration Region

$$F_Z(z) = \iint_{x+y \leqslant z} f_{XY}(x, y)dxdy$$

Then

$$f_Z(z) = \frac{d}{dz}F_Z(z)$$

Need derivative of integral with variable limits.

(page 7)

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7. Leibnitz Rule (important theorem)

Slides give:

If

$$G(x) = \int_{a(x)}^{b(x)} h(x, y)dy$$

Then

$$\frac{d}{dx}G(x) = h(x, b(x))b^{'}(x) - h(x, a(x))a^{'}(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x}h(x, y)dy$$

(page 8)

Reasoning:

Derivative of integral must consider:

1. upper limit change
2. lower limit change
3. integrand change

Used for transformation of 2 RVs.

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8. Derivation for Z = X + Y

Start:

$$F_Z(z) = P(X + Y \leqslant z)$$

Integrate joint PDF.

After differentiation:

$$f_Z(z) = \int f_{XY}(x, z - x)dx$$

(page 9–10)

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9. Independent Case → Convolution

If X and Y independent:

$$f_{XY}(x, y) = f_X(x)f_Y(y)$$

Then

$$f_Z(z) = \int f_X(x)f_Y(z - x)dx$$

Slides label this:

> Convolution integral

(page 11)

Reasoning:

Joint PDF factorizes → simpler formula

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10. Example — Normal + Normal

Given:

$$X \sim N(0, 1)$$

$$Y \sim N(0, 1)$$

Independent

Then

$$Z = X + Y$$

Use convolution.

After integration:

$$Z \sim N(0, 2)$$

(page 10–12)

Reasoning:

Variance adds:

$$\sigma_Z^2 = \sigma_X^2 + \sigma_Y^2 = 1 + 1 = 2$$

---

11. Example — Exponential + Exponential

Given:

$$f_X(x) = \lambda e^{-\lambda x}, x > 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, y > 0$$

Independent

Use convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Simplify:

$$f_Z(z) = \lambda^2 z e^{-\lambda z}$$

for

$$z > 0$$

(page 13)

Reasoning:

Limits from support of exponential.

12. Example — Z = X − Y

Use same method.

$$F_Z(z) = P(X - Y \leqslant z)$$

Region changes.

Integrate over region.

Differentiate.

(page 14)

Reasoning:

Always:

1. Write CDF
2. Find region
3. Integrate
4. Differentiate

---

13. Example — Ratio

$$Z = X/Y$$

Need to consider:

- Y > 0
- Y < 0

Limits change.

(page 15)

Reasoning:

Division changes inequality direction.

---

14. Example — Polar Transformation

Given:

$$R = \sqrt{X^2 + Y^2}$$

Need transformation method.

Use joint PDF + change of variables.

(page 15)

Final Exam Notes (Important)

✔️ Always start from CDF
✔️ Use inverse transformation for 1 RV
✔️ Use joint PDF for 2 RV
✔️ Use Leibnitz rule for derivative
✔️ Independent → convolution
✔️ Normal + Normal → variance adds
✔️ Exponential + Exponential → Gamma
✔️ Region of integration is key

If you want, I can also make **formula sheet for Lecture 12 (1-page revision)**.

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Below is a **faithful, step-by-step lecture scribe for Lecture 12 — Transformation of Random Variables**, written strictly from the provided slides, with **explicit reasoning included inside the scribe**, organized exactly in lecture order so a student can **reconstruct definitions, derivations, and examples during a closed-notes exam**. All content comes only from the slides.

Lecture 12 Scribe

Transformation of Random Variables + Function of Two Random Variables + Example

---

1. Lecture Objective and Outline

The lecture states three topics:

1. **Transformation of Random Variables**
   Learn how to obtain the PDF of a new random variable after applying a function.
2. **Function of Two Random Variables**
   Study joint transformations and derived distributions.

3. **Illustrative Example**
   Detailed derivation for the case

$$Z = X + Y$$

Important assumption written in slides:

- PDF of original random variable is known.
- Goal: find PDF of transformed random variable.

Reasoning:
If we know the distribution of X and apply a function, we must compute the distribution of the new variable using probability rules.
(page 2)

---

2. Transformation of One Random Variable

Let

$$Y = g(X)$$

Goal:

$$f_Y(y) = ?$$

Slides show the method using **CDF first**.

Step 1 — Write CDF

$$F_Y(y) = P(Y \leqslant y)$$

Substitute $Y = g(X)$

$$F_Y(y) = P(g(X) \leqslant y)$$

Reasoning:
PDF is hard to transform directly, but CDF uses inequalities, which are easier to manipulate.

Step 2 — Use inverse function

If $g(x)$ is monotonic increasing,

$$P(g(X) \leqslant y) = P(X \leqslant g^{-1}(y))$$

So,

$$F_Y(y) = F_X(g^{-1}(y))$$

Reasoning:
Monotonic function allows inequality to be inverted.

Step 3 — Differentiate

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \frac{dx}{dy}$$

Final formula in slide:

$$f_Y(y) = \frac{f_X(x)}{\left|\frac{dy}{dx}\right|} \quad x = g^{-1}(y)$$

Reasoning:

- derivative appears due to change of variable
- absolute value because derivative can be negative

(page 3)

---

3. Case when function is decreasing

Slides show second case.

If g is decreasing:

$$F_Y(y) = P(Y \leqslant y) = P(X \geqslant g^{-1}(y))$$
$$= 1 - F_X(g^{-1}(y))$$

Differentiate:

Same final result

$$f_Y(y) = \frac{f_X(x)}{\left|\frac{dy}{dx}\right|}$$

Reasoning:

Inequality reverses, but derivative formula remains same.

(page 4)

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4. Example — Nonlinear Transformation

Given in slide:

$$X \sim U(-1, 1)$$

$$f_X(x) = \begin{cases} 1/2, & -1 < x < 1 \\ 0, & otherwise \end{cases}$$

Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Find $f_Y(y)$

Step 1 — Find inverse

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Reasoning:
Need inverse to use transformation formula.

Step 2 — derivative

$$\frac{dx}{dy} = \frac{2}{\pi}\frac{1}{\sqrt{1-y^2}}$$

Step 3 — apply formula

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute

$$= \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1 - y^2}}$$

$$= \frac{1}{\pi \sqrt{1 - y^2}}$$

for

$$-1 < y < 1$$

(page 5)

Reasoning:

- Use inverse
- Compute derivative
- Multiply by original PDF

---

5. Function of Two Random Variables

Now consider

$$Z = X + Y$$

Goal:

$$f_Z(z)$$

Slides say:

Need joint PDF

$$f_{XY}(x, y)$$

Reasoning:

One variable → use derivative
Two variables → need integration

(page 6)

6. Use CDF method

$$F_Z(z) = P(Z \leqslant z) = P(X + Y \leqslant z)$$

Write as integral over region

$$F_Z(z) = \iint_{x+y \leqslant z} f_{XY}(x, y)dxdy$$

Then

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

(page 7)

Reasoning:

Region depends on z → limits depend on z → need Leibnitz rule.

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7. Leibnitz Rule (given in slide)

If

$$G(x) = \int_{a(x)}^{b(x)} h(x, y)dy$$

Then

$$\frac{d}{dx} G(x) = h(x, b(x))b'(x) - h(x, a(x))a'(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y)dy$$

Reasoning:

Derivative must consider

1. upper limit
2. lower limit
3. integrand

(page 8)

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8. Derivation for Z = X + Y

Start

$$F_Z(z) = P(X + Y \leqslant z)$$

Integrate joint PDF.

After differentiation (shown in slides):

$$f_Z(z) = \int f_{XY}(x, z - x)dx$$

(page 9)

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9. Independent case → convolution

If independent:

$$f_{XY}(x, y) = f_X(x)f_Y(y)$$

Then

$$f_Z(z) = \int f_X(x)f_Y(z - x)dx$$

Slides label:

Convolution integral

(page 11)

Reasoning:

Joint PDF becomes product.

---

10. Example — Normal + Normal

Given:

$$X \sim N(0, 1)$$

$$Y \sim N(0, 1)$$

Independent

Then

$$Z = X + Y$$

Using convolution → slides show result

$$Z \sim N(0, 2)$$

Reasoning:

Variance adds.

(page 10–12)

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11. Example — Exponential + Exponential

Given

$$f_X(x) = \lambda e^{-\lambda x}, x > 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, y > 0$$

Independent

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Slides result:

$$f_Z(z) = \lambda^2 z e^{-\lambda z}$$

for $z > 0$

(page 13)

Reasoning:

Limits from support of exponential.

---

12. Example — Z = X − Y

Slides show:

$$F_Z(z) = P(X - Y \leqslant z)$$

Integrate over region

Differentiate

Same method as Z = X+Y

(page 14)

---

13. Example — Ratio

$$Z = X/Y$$

Need to consider

- Y > 0
- Y < 0

Because inequality changes.

(page 15)

Reasoning:

Division changes sign of inequality.

---

14. Example — Radius

Slides show

$$R = \sqrt{X^2 + Y^2}$$

Need transformation of two variables.

(page 15)

Reasoning:

Use joint PDF and change of variables.

Final Exam Summary (from lecture flow)

✔️ Use CDF method for transformation
✔️ For one RV → inverse + derivative
✔️ For two RV → joint PDF + integration
✔️ Use Leibnitz rule
✔️ Independent → convolution
✔️ Normal + Normal → variance adds
✔️ Exponential + Exponential → convolution
✔️ Region of integration must be correct

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All statements are taken only from the slides.

Lecture 12 Scribe

Transformation of Random Variables + Function of Two Random Variables + Example

---

1. Lecture Outline and Objective

Topics listed in the slides:

1. **Transformation of Random Variables**
   Learning techniques to find the PDF of a transformed random variable.
2. **Function of Two Random Variables**
   Joint transformations and derived distributions.
3. **Illustrative Example**
   Detailed derivation for the case

$$Z = X + Y$$

Important assumption written in slide:

- PDF of the original random variable is known.
- Goal: find PDF of the new transformed random variable.

Reasoning given in lecture flow:

We know the distribution of X, but after applying a function, the distribution changes.

Therefore we must derive the new PDF using probability definitions.
(page 2)

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2. Transformation of One Random Variable

Let

$$Y = g(X)$$

Goal:

$$f_Y(y)$$

Slides show that the correct method is:

1. Find CDF first
2. Differentiate to get PDF

Reasoning:

PDF cannot be transformed directly, but CDF uses inequalities which are easier to manipulate.

Step 1 — Write CDF

$$F_Y(y) = P(Y \leqslant y)$$

Substitute $Y = g(X)$

$$F_Y(y) = P(g(X) \leqslant y)$$

Step 2 — Use inverse function

If $g(x)$ is monotonic increasing,

$$P(g(X) \leqslant y) = P(X \leqslant g^{-1}(y))$$

So,

$$F_Y(y) = F_X(g^{-1}(y))$$

Reasoning:

Monotonic function allows us to invert inequality.

(page 3)

Step 3 — Differentiate

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \frac{dx}{dy}$$

Slides give final result:

$$f_Y(y) = \frac{f_X(x)}{\left|\frac{dy}{dx}\right|} \quad x = g^{-1}(y)$$

Reasoning:

- derivative appears due to change of variable
- absolute value needed because derivative may be negative

(page 3–4)

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3. Case when function is decreasing

Slides show second case.

If $g(x)$ is decreasing:

$$F_Y(y) = P(Y \leqslant y) = P(X \geqslant g^{-1}(y))$$
$$= 1 - F_X(g^{-1}(y))$$

Differentiate → same final formula

$$f_Y(y) = \frac{f_X(x)}{\left|\frac{dy}{dx}\right|}$$

Reasoning:

Inequality reverses but derivative formula stays same.

(page 4)

---

4. Example — Nonlinear Transformation

Given in slide:

$$X \sim U(-1, 1)$$

$$f_X(x) = \begin{cases} 1/2, & -1 < x < 1 \\ 0, & otherwise \end{cases}$$

Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Find $f_Y(y)$

Step 1 — Find inverse

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Reasoning:

Need inverse to substitute into formula.

Step 2 — derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1 - y^2}}$$

Step 3 — Apply formula

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute:

$$= \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

$$= \frac{1}{\pi\sqrt{1-y^2}}$$

for

$$-1 < y < 1$$

(page 5)

Reasoning:

1. invert
2. differentiate
3. multiply by original PDF

---

5. Function of Two Random Variables

Now consider

$$Z = X + Y$$

Goal:

$$f_Z(z)$$

Slides state:

Need joint PDF

$$f_{XY}(x, y)$$

Reasoning:

With two variables, transformation depends on both, so integration is required.

(page 6)

---

6. Use CDF method

$$F_Z(z) = P(Z \leqslant z) = P(X + Y \leqslant z)$$

Write as integral:

$$F_Z(z) = \iint_{x+y \leqslant z} f_{XY}(x, y)dxdy$$

Then

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Reasoning:

Limits depend on z → need derivative of integral.

(page 7)

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7. Leibnitz Rule

Slides give rule:

If

$$G(x) = \int_{a(x)}^{b(x)} h(x, y)dy$$

Then

$$\frac{d}{dx} G(x) = h(x, b(x))b^{'}(x) - h(x, a(x))a^{'}(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y)dy$$

Reasoning:

Derivative must consider

1. upper limit
2. lower limit
3. integrand

(page 8)

---

8. Result for Z = X + Y

Slides derive:

$$f_Z(z) = \int f_{XY}(x, z - x)dx$$

(page 9)

Reasoning:

After applying Leibnitz rule.

---

9. Independent case → Convolution

If X and Y independent:

$$f_{XY}(x, y) = f_X(x)f_Y(y)$$

Then

$$f_Z(z) = \int f_X(x)f_Y(z - x)dx$$

Slides label:

Convolution integral

(page 11)

---

10. Example — Normal + Normal

Given:

$$X \sim N(0, 1)$$

$$Y \sim N(0, 1)$$

Independent

Then

$$Z = X + Y$$

Slides result:

$$Z \sim N(0, 2)$$

Reasoning shown in slides:

Variance adds.

(page 10–12)

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11. Example — Exponential + Exponential

Given:

$$f_X(x) = \lambda e^{-\lambda x}, x > 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, y > 0$$

Independent

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Slides result:

$$f_Z(z) = \lambda^2 z e^{-\lambda z}$$

(page 13)

---

12. Example — Z = X − Y

Slides:

$$F_Z(z) = P(X - Y \leqslant z)$$

Integrate region

Differentiate

Same method as Z=X+Y.

(page 14)

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13. Example — Ratio

$$Z = X/Y$$

Need to consider

- Y>0
- Y<0

because inequality changes.

(page 15)

---

14. Example — Radius

$$R = \sqrt{X^2 + Y^2}$$

Need transformation of two variables.

(page 15)

Final Exam Reconstruction Steps

From lecture flow:

1. Write transformation
2. Use CDF method
3. Find limits
4. Differentiate
5. Use Leibnitz rule if needed
6. If independent → convolution
7. Check support limits

These steps match the order used in slides.

Do you like this personality?